

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

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Sequential strain and doping scans are standard in covalent molecular networks, but total-energy effects need not add. In quasi-hexagonal C₆₀ graphullerene we quantify the omitted cross term with a synergy order parameter $\mathcal{S}$ (meV/atom). PBE+DFT-D3 on tetramers and periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$) gives opposite strain coefficients for N (−303 meV/%) and B (+57 meV/%) and $|\mathcal{S}|$ up to ~ 32 meV/atom at +3% strain, largest for P despite small $|\alpha|$. The pattern warns that sequential high-throughput screens can mis-rank stability when local dopant distortion couples nonlinearly to external strain.

I. INTRODUCTION

Covalent molecular networks—from graphullerenes to covalent organic frameworks—are often tuned jointly by substitutional chemistry and mechanical load, yet many pipelines still estimate $E_{\text{tot}}(\epsilon, \delta)$ from *separate* doping-only and strain-only calculations [1, 2]. Synthetic carbon allotropes with extended π conjugation underpin optoelectronic applications, yet charge transport in molecular networks is often polaronic [3, 4]. Quasi-hexagonal (qHP) C₆₀ graphullerene [3, 4] provides a covalently linked model network for such coupled tuning, with interfullerene linking that raises hopping couplings relative to vdW C₆₀ monolayers [3]. Beyond pristine monolayers, strain and substitution independently reshape band edges [2, 5, 6], but high-throughput pipelines often compose $E_{\text{tot}}(\epsilon, \delta)$ from *separate* scans. Non-additive coupling is documented in other two-dimensional networks [1]; a quantitative *total-energy* cross term $\mathcal{S}$ [Eq. (1)] has not been reported for substitutional B/N/P qHP C₆₀ tetramers and $n = 1\text{--}8$ supercells under a single PBE+D3 protocol.

We address three questions on this model network: (1) how opposite linear strain coefficients α emerge for donor vs. acceptor dopants; (2) how large the omitted cross term $|\mathcal{S}|$ is when strain and substitution are applied jointly; and (3) whether $|\mathcal{S}|$ tracks $|\alpha|$ or instead reflects local structural distortion. Using $\gtrsim 90$ converged CP2K tasks [7] (Table S2), we define $\mathcal{S}$ on shared tetramer and periodic grids; Marcus λ and hopping observables (IPR/ J) are analyzed only in Supplemental Material Fig. S5–S6 and are not required for the non-additivity conclusions here. The central narrative anchor is that *local* dopant distortion and *global* biaxial strain interact nonlinearly in total energy: $|\mathcal{S}|$ can exceed what linear α alone would suggest, so stability rankings from $\epsilon = 0$ substitution energies or from strain-only elastic increments need not survive a coupled load.

II. COMPUTATIONAL METHODS

A. Electronic-structure setup

All calculations use CP2K [8] with PBE [9] and Grimme DFT-D3 (Becke–Johnson damping) [10] (Supplemental Material for basis, SCF, and task inventory). We deliberately remain below the Koopmans/rVV10 level of Ref. 3 to obtain a uniform, size-converged (ϵ, δ) scan rather than absolute polaron binding energies.

B. Notation and conventions

Biaxial strain is denoted ϵ (percent); dopant species $\delta \in \{B, N, P\}$. The synergy order parameter is always $\mathcal{S}$ (calligraphic) in meV/atom [Eq. (1)]. π -projected densities of states are written π -DOS (never bare “pi” in equations).

Rigid-strain protocol. Biaxial strain scales the in-plane lattice vectors of the *pristine* equilibrium cell at fixed fractional coordinates (no ionic relaxation under load). Reported α and $\mathcal{S}$ under this protocol are *upper bounds* on coupling under load until the ionic-relaxation benchmark in Table S3 converges.

Plane-wave cutoff (size scaling). Archived CP2K inputs use 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (Table S2); comparisons at fixed n always share the same cutoff. Cross- n trends therefore mix a modest basis change at $n = 6$; we do *not* interpret the reported N $\mathcal{S}$ sign change between $n = 6$ and $n = 8$ as a physical size effect until a matched 400 Ry $n = 6$ control completes (input `size_6x60_N_pos3pct_cutoff400`).

C. Substitutional doping and grids

Tetramer models ($4 \times \text{C}_{60}$, 240 sites) substitute B, N, or P at nominal 5% (12–13 sites from archived coordinates, random seed 42), not a thermodynamic minimum. A second seed 137 tetramer ensemble (Table S4) tests whether α and tetramer-scale $\mathcal{S}$ are sensitive to dopant placement;

periodic $\mathcal{S}(n)$ claims remain on the seed 42 periodic protocol until seed 137 periodic points are added. Periodic scaling cells place one dopant per C_{60} ($n = 1-8$), so nominal concentration falls from $\sim 5\%$ to $\sim 1.7\%$ per atom at $n = 4$. Tetramer α , gaps, and Supplemental Material IPR/ J trends are therefore not directly interchangeable with periodic $\mathcal{S}(n)$ without a matched-concentration benchmark; all $\mathcal{S}$ claims use the periodic protocol. In total, $\gtrsim 90$ converged neutral tasks support the quantitative claims below (Table S2). Charged polaron geometry optimizations and vertical single-points for Marcus reorganization energies follow the same functional and basis choices but are reported separately in Supplemental Material Fig. S5 once converged; they do not enter the $\mathcal{S}$ definition in Eq. (1).

D. Energy observables

Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain, tetramer *total* energy) is obtained from the 24-point grid (Table S1). Positive α means tension *raises* total energy; negative α means tension *lowers* it under the rigid-strain protocol. Substitution energy $E_f = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is reported relative to pristine tetramers at the same strain (no chemical-potential reference). In sequential high-throughput screens one often estimates $E_{\text{tot}}(\epsilon, \delta) \approx E_f(\epsilon=0) + E_{\text{strain}}(\epsilon)$; the neglected term is precisely $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. For screening workflows, the magnitude of this error is best gauged against the substitution-energy scale: at $n = 4$ and +3% strain, P has $|\mathcal{S}| \approx 24$ meV/atom while the per-atom substitution scale is $|E_f| \times n_{\text{dop}}/n_{\text{atoms}} \approx 0.79$ eV/atom ($|E_f| \approx 16$ eV per P site, $n_{\text{dop}}=12$, Table S1)—i.e. $|\mathcal{S}|/|E_f|_{\text{per atom}} \sim 3\%$, large enough to reorder stability margins in sequential doping-then-strain screens even though $|\mathcal{S}|$ exceeds the rigid per-atom strain increment ($\alpha \times 3\%/N_{\text{atoms}} \lesssim 0.3$ meV/atom for P).

Non-additive coupling metric. We quantify strain-doping cross terms with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times C_{60}$. The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ indicates cooperative stabilization beyond the two-scan sum, whereas $\mathcal{S} > 0$ indicates cooperative destabilization.

To first order in strain, $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon$ when $\alpha_\delta = dE/d\epsilon$ is evaluated for the doped cell and α_0 for the pristine reference at the same n (rigid-strain protocol); this links the mechanical coefficient α to the cross term at small ϵ .

Periodic-cell $\mathcal{S}$ values are obtained from archived total energies via post-processing of converged CP2K outputs (Supplemental Material Table S2). We report $\mathcal{S}$ for total energy only; an analogous gap cross term $\mathcal{S}_{\text{gap}}$ is not computed here—main-text panel (a) shows that gap slopes need not track $|\mathcal{S}|$.

Localization and coupling (Supplemental Material Fig. S6). Following Capobianco *et al.* [3], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling J from tetramer dimer fragments. IPR follows Supporting Information Eq. (S8), $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$, on a Mulliken-projected orbital grid; J is estimated from the ground-first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model). These observables are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant $J \simeq 75-81$ meV reported for pristine qHP networks [3], but they provide an internally consistent trend under coupled (ϵ, δ) .

Polaron binding (Supplemental Material). Marcus reorganization energies follow the vertical-adiabatic decomposition [3]: $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$, with separate internal ($\lambda_{\text{loc}}^\pm$) and environmental ($\lambda_{\text{env}}^\pm$) contributions when geometry subsets are relaxed. Charged-state relaxations and vertical single-points are documented in Supplemental Material Fig. S5.

E. Validation benchmarks (Supplemental Material)

(i) Ionic relaxation (Table S3). Four fixed-cell GEO.OPT corners (pristine and P-doped at $\epsilon = 0, +3\%$) quantify how much rigid $\mathcal{S}$ overestimates the cross term when atoms relax under the same biaxial cell. **(ii) Alternate dopant ensemble (Table S4).** Tetramer B/N/P at seed 137 repeats the $\epsilon = 0, +3\%$ rigid grid to bound placement sensitivity. **(iii) Cutoff control (Table S2).** A single $6 \times C_{60}$ N @ +3% task at 400 Ry isolates basis-set confounding from electrostatic size trends.

III. RESULTS

We organize the results around total-energy non-additivity: Sec. III A summarizes band-edge context; Sec. III B reports linear strain coefficients and E_f ; Sec. III C gives $\mathcal{S}(n)$ at +3% strain across dopants and supercell sizes (IPR/ J and Marcus λ remain in Supplemental Material Fig. S5–S6).

A. Electronic structure under strain and doping

Figure 1(a) reports HOMO–LUMO gaps from tetramer single-points at $\epsilon \in \{-5, 0, +5\}\%$. B (acceptor) introduces states near the valence manifold and compresses the π gap toward zero at $\epsilon = 0$; N (donor) pushes

electron density into the conduction manifold; P perturbs the network more heterogeneously because of its larger covalent radius and longer cage bonds. Tensile strain generally narrows the pristine gap while dopant-specific slopes differ: B gaps remain the smallest across the grid, whereas N and P retain finite gaps at $\epsilon = 0$ (panel (a)). Figure 1(b) shows representative π -DOS for B @ $\epsilon = 0$ (valence-edge and conduction-edge coloring follows Ref. 3). These band-edge trends contextualize the mechanical and non-additive energy results but are not used to infer $\mathcal{S}$.

B. Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%). Substitution opens an order-of-magnitude spread in α : N (-303.2 meV/%, tension stabilizes the doped cell), B ($+56.9$ meV/%, tension destabilizes), and P (-19.1 meV/%, weakly stabilizing) (Table S1). The opposite sign for N vs. B persists across the $\epsilon \in \{-5, \dots, +5\}$ % grid and is not an artifact of a single strain point. At $\epsilon = +5\%$, N remains the most tension-stabilized dopant (α most negative), whereas B shows the strongest destabilization under tension; P sits between these extremes with a modest negative α , foreshadowing why its linear coefficient poorly predicts the large $|\mathcal{S}|$ at $+3\%$ in periodic cells. Substitution energies at $\epsilon = 0$ span $E_f = +73.4$ eV/dopant (B), -115.3 eV/dopant (N), and -15.8 eV/dopant (P), with $n_{\text{dop}} = 12$ –13 counted from input coordinates (Table S1). Figure 1(c) highlights the α split; ranking dopants by $\epsilon = 0$ $|E_f|$ alone is misleading once $+3\%$ strain is applied.

C. Non-additive strain–doping coupling

Figure 1(d) tracks $\mathcal{S}(n)$ with one dopant per C_{60} at $+3\%$ biaxial strain. At $n = 4$, $\mathcal{S} = -6.2$ meV/atom (B), $+3.0$ meV/atom (N), and -23.7 meV/atom (P); P shows the largest magnitude despite the smallest $|\alpha|$ on the tetramer grid (Table S1). Across fifteen coupled points ($n \in \{1, 2, 4, 6, 8\}$, B/N/P), $|\mathcal{S}|$ reaches ~ 32 meV/atom for P at $n = 1$ and remains $\mathcal{O}(10)$ meV/atom at larger n for B and P, whereas N is smaller in magnitude at $n = 4$ but comparable at $n = 1$. For N, $\mathcal{S}$ changes sign between $n = 6$ ($+2.4$ meV/atom) and $n = 8$ (-2.3 meV/atom) at the production 350 Ry cutoff; this reversal is *not* used in the main conclusions until a matched 400 Ry $n = 6$ control and Table S3–S4 benchmarks complete (Sec. II E). Size trends for B and P at fixed cutoff are more monotonic: $|\mathcal{S}|$ decreases from the $n = 1$ corner toward $n = 4$ but remains $\mathcal{O}(10)$ meV/atom at $n = 4$ –8. We report provisional $\mathcal{S}(n) \approx \mathcal{S}_\infty + A/n$ fits in Supplemental Material but withhold asymptotic $\mathcal{S}_\infty$ claims until cutoffs are matched across n . Negative $\mathcal{S}$ (B, P at $n = 4$) means coupled tension *stabilizes* the doped cell relative to the

additive estimate; positive $\mathcal{S}$ (N at $n = 4$) means sequential screens *overestimate* stability under load.

IV. DISCUSSION

Context. Capobianco *et al.* [3] established qHP C_{60} as a polaronic network whose mobility gain over vdW C_{60} tracks enhanced interfullerene coupling rather than reduced reorganization energy. Li *et al.* [2] and Khan and Leuenberger [5] show that strain and impurity engineering independently move band edges in fullerene monolayers, while Materials2024untangling reports non-additive strain–doping effects in other 2D networks. Our contribution is complementary: at fixed PBE+D3, we quantify the *total-energy* cross term $\mathcal{S}$ omitted when those knobs are applied sequentially, distinct from hopping metrics analyzed only in Supplemental Material Fig. S6. Where Capobianco *et al.* [3] emphasize that mobility gains track interfullerene J rather than reduced λ , our results show that *total-energy* stability under simultaneous strain and substitution is likewise not reducible to a sum of one-parameter scans—a caution that applies to materials discovery pipelines even when transport figures of merit are not computed.

Mechanistic synthesis (three coupling classes).

(i) Opposite α for N vs. B. N introduces excess π electrons that strengthen bonding under tension (negative α), whereas B creates electron deficiency and antibonding π^* character that rises with lattice expansion (positive α)—consistent with donor vs. acceptor π -counting and the PDOS shifts in Fig. 1(a,b). **(ii) Large $|\mathcal{S}|$ for P despite small $|\alpha|$.** P’s covalent radius (~ 107 pm vs. ~ 77 pm for C) and longer P–C bonds impose intrinsic cage distortion before external load is applied; tensile strain then couples to this pre-existing local stress field rather than to a pristine cage, yielding a cross term that is invisible in separate doping-only and strain-only scans but dominant when both are engaged—consistent with P’s intermediate E_f yet largest $|\mathcal{S}|$ at $+3\%$ and with the weak tetramer $|\alpha|$. **(iii) N $\mathcal{S}$ sign change with n .** With one dopant per C_{60} , electrostatic fields dilute at large n ; any N sign flip between $n = 6$ and $n = 8$ is confounded with the 400→350 Ry cutoff step and remains uninterpreted (Sec. II E). Together, (i)–(iii) suggest a taxonomy for covalent molecular networks: *electronic* donor/acceptor control of α , *structural* local-distortion coupling to external strain (P), and *finite-size* electrostatic dilution (N at large n).

(iv) Linear $\mathcal{S}$ – α check. Using tetramer α (Table S1) in $\mathcal{S} \approx (\alpha_\delta - \alpha_0)\epsilon$ at $\epsilon = +3\%$ and dividing by 240 sites gives $\sim +0.7$, -3.8 , and -0.2 meV/atom for B/N/P—far smaller than the periodic $n=4$ values (-6.2 , $+3.0$, -23.7 meV/atom). The gap shows that tetramer-scale linear mechanics and 5% doping do not quantitatively predict periodic $\mathcal{S}$; for P, nonlinear coupling dominates beyond first-order $(\alpha_\delta - \alpha_0)\epsilon$.

(v) Significance versus DFT uncertainty. At

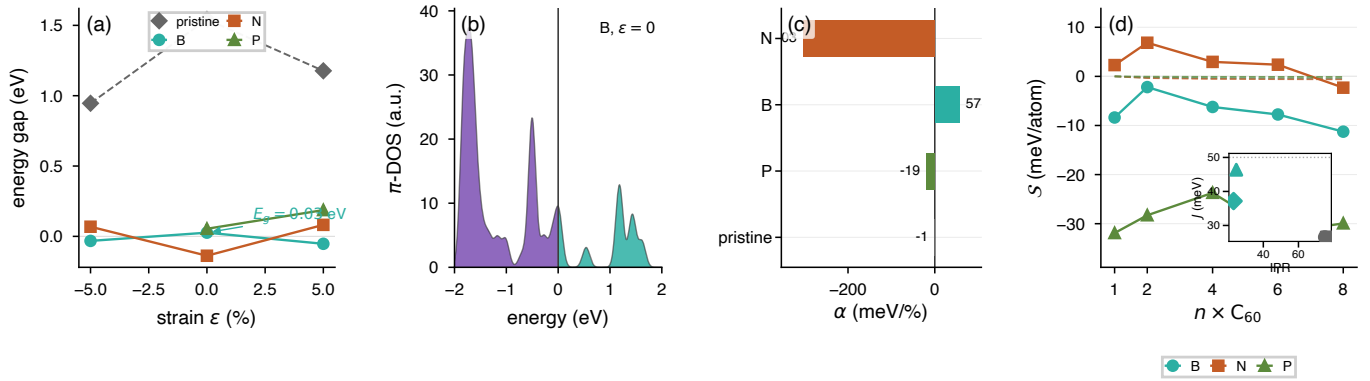


FIG. 1. Coupled (ϵ, δ) response in doped qHP C_{60} (PBE+D3). (a) HOMO-LUMO gap vs. strain. (b) π -DOS for B @ $\epsilon=0$ (Electron SI palette). (c) Mechanical α . (d) $S(n)$ at +3%; inset: schematic IPR/ J grid (details in Supplemental Material Fig. S6).

$n=4$ and +3% strain, $|S| \sim 24$ meV/atom is $\sim 3\%$ of the per-atom substitution-energy scale (Table S1) but still $\gg$ typical rigid per-atom increments from α . PBE formation energies for molecular solids can carry $\mathcal{O}(10^2)$ meV/atom uncertainties; we therefore treat S as a *coupled-scan correction* to sequential workflows rather than an absolute formation-energy prediction, and flag cases where $|S|$ exceeds the rigid strain increment as the primary screening hazard.

Design implications. At +3% strain, P carries the largest additive-approximation error in S ($|S| \sim 24$ – 32 meV/atom); co-design should treat P under tension as a *coupled* stability problem, not $|E_f|$ at $\epsilon = 0$ alone. Where $S < 0$ (B and P at $n=4$), coupled tension can stabilize doped cells beyond additive predictions; where $S > 0$ (N at $n=4$), sequential screens overestimate stability under load. For example, a workflow that ranks P vs. B stability at $\epsilon = 0$ from E_f alone and then applies +3% strain sequentially would miss a ~ 24 meV/atom correction for P—comparable to several percent of the substitution-energy scale and much larger than the rigid strain increment from α . Gap closing [Fig. 1(a)] does not track $|S|$; we do not compute S_{gap} , so photonic or transport targets cannot be inferred from total-energy S alone. For B at $n = 4$, both negative S and positive α imply that a strain-only elastic estimate would predict tension *raises* energy while the coupled scan *lowers* it relative to the additive model—a direct sign conflict that sequential pipelines cannot resolve without evaluating S . N S vs. n is reported for completeness but excluded from design rules until Table S2–S4 controls converge.

Limitations and outlook. Three protocol limits are actively benchmarked (Sec. II E): rigid strain (Table S3), fixed-seed ensembles (Table S4), and mixed 400/350 Ry cutoffs across n (Table S2 control task). Additional limits: tetramer vs. periodic concentration mismatch; PBE+D3 without Koopmans/rVV10; S_{gap} not quantified; Marcus λ pending charged vertical SP (Supplemen-

tal Material Fig. S5).

V. CONCLUSIONS

In discrete covalent molecular networks, nonlinear coupling between dopant-induced local distortion and external strain can dominate sequential strain-doping energy estimates. Using qHP C_{60} as a model system, we quantified this cross term with S (meV/atom): $|S|$ is not simply tied to linear α (P shows the largest magnitude despite small $|\alpha|$), and additive scans can mis-rank stability at the few-percent level of the substitution-energy scale. Opposite N vs. B strain coefficients and $|S|$ at $n = 4$ for B and P show that donor/acceptor chemistry and load must be co-designed; size-dependent N trends await cutoff and placement controls (Tables S2–S4). Ionic-relaxation, alternate-seed, and matched-cutoff benchmarks remain to test generality beyond the rigid seed 42 protocol applied here.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The data that support the findings of this study are openly available in the `sci-simukit` repository at <https://github.com/chenxingqiang/graphullerene-strain-engineering>, including converged CP2K inputs and outputs under `dft_results/`, analysis JSON under `experiments/analysis/`, and figure-generation scripts under `paper/figures/`. A tagged release matching the manuscript revision will be issued upon acceptance.

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